NAME: ______________________ **PERIOD:** ________

Two Ways to Sample Library Visits

AP Statistics · Unit 3 · Topic 3.9

[STAR] THE PROBLEM

A library studies self-checkout before and after a redesign. Records give $p_B = 0.42$ among 4,000 before visits and $p_A = 0.54$ among 5,000 after visits.

Plan I: Draw an SRS of 120 before visits and a separate SRS of 150 after visits; do not track cardholders.

Plan P: Measure the same 120 cardholders before and after.

Sora: Separate selection warrants Plan I's independent model; 0.0350 means a difference at most 0.01 is unusual.

Mateo: Pairing controls person differences, so Plan P may help. Because both proportions use 120 people, Plan I's model applies to P too.

For Plan I, the four expected counts are 81, 69, 50.4, and 69.6. A normal calculator gives the lower-tail probability 0.0350.

Sampling plans

Plan	Before	After	Repeated
I	SRS 120	SRS 150	no
P	120	same 120	yes

FIRST TAKE — one sentence before you work the parts

Does using the same people twice seem different from choosing two separate groups? Give one reason.

[BOOK] THE VARIANCE FORMULA REQUIRES INDEPENDENT SAMPLES (keep on the page)

For independent samples, $\mu_{\widehat{p}_1-\widehat{p}_2} = p_1 - p_2$ and $\sigma_{\widehat{p}_1-\widehat{p}_2} = \sqrt{p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2}$ when each sample is below 10% of its population. Approximate normality requires all four expected counts to be at least 10. Repeated measurements on the same people are paired, not independent; their association belongs in the variability.

Work (a)–(c).

DOK 1 **(a)** For after minus before, identify the parameter and statistic and state Plan I's sampling-distribution mean.

DOK 2 **(b)** For Plan I, verify both 10% conditions and use the four supplied counts to check normality. Calculate the SD and interpret the supplied probability $P(\widehat{p}_A - \widehat{p}_B \leq 0.01) = 0.0350$.

DOK 3 **(c)** In 3–4 sentences, explain what Sora and Mateo each get right and wrong. Use how the samples are selected to decide where the independent model applies. Explain why transferring its SD and probability to the repeated people could mislead a reader.

Frame: Plan I supports the independent model because ____________________.

Frame: In Plan P, the responses are ____________; transferring the formula could ____________________.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.